AN ETALE EQUIVALENCE RELATION ON AN AFFINE SCHEME HAS AN ALGEBRAIC SPACE QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This candidate was dropped by the mechanical prefilter because the phrase 'This is clear from the above' occurs once at proof line 1749; the read shows it introduces a short recap of the first step inside a 126-line complete proof, not an omission.

Editorial note. The proof contains an eqnarray, two displays and one xymatrix cube; all are part of the counted proof span and are preserved verbatim, with the chapter's own Xy-pic setup retained in the preamble.

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- A monomorphism of schemes is separated. Schemes, Lemma schemes-lemma-monomorphism-separated, used at proof lines 1692-1694. Makes j separated.
- Permanence of finite type and finite fibres. Morphisms, Lemmas morphisms-lemma-permanence-finite-type and morphisms-lemma-finite-fibre, used at proof lines 1698-1704. Make j locally quasi-finite.
- Descent data for sheaves and their effectiveness. Descent, Lemma descent-lemma-descent-data-sheaves, used at proof lines 1716-1725 and 1800-1817. Turns the local models X_i -> T_i and X -> T' into representability of G.
- Separated locally quasi-finite morphisms descend through fppf coverings. More on Morphisms, Lemma more-morphisms-lemma-separated-locally-quasi-finite-morphisms-fppf-descend, used at proof lines 1721-1725 and 1813-1817. Makes the descent datum effective.
- Unramified or flat and unramified morphisms are etale. Morphisms, Lemmas morphisms-lemma-unramified-quasi-finite and morphisms-lemma-flat-unramified-etale, used at proof lines 1712-1715. Supplies the properties of t used in the descent.
- Products of representable transformations and their properties. Lemmas lemma-product-representable-transformations (409-420, produced as L44195) and lemma-product-representable-transformations-property, used at proof lines 1763-1774. Make U x_S U -> F x F representable, surjective and etale.

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- Locality on the base for the relevant properties. Remark remark-list-properties-fpqc-local-base, used at proof lines 1753-1756 and 1773-1775. Lifts surjectivity and etaleness from the covering.

Source reference. Original lemma 02W5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3792–3792. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3798–3798. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3797–3797. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3817–3817. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02W9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3796–3796. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3799–3799. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3809–3809. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3818–3818. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02W8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3795–3795. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02NG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3479–3479. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02GV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3242–3242. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01T8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2365–2365. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02V5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3756–3756. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 02WH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3804–3804. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 02WG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3803–3803. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01L4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2073–2073. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let U be a scheme over S. Let $j = (s, t) : R \to U \times_S U$ be an étale equivalence relation on U over S. Assume that U is affine. Then the quotient $F = U/R$ is an algebraic space, and $U \to F$ is étale and surjective.*

Proof. Since $j : R \to U \times_S U$ is a monomorphism we see that j is separated (see Schemes, Lemma 01L4). Since U is affine we see that $U \times_S U$ (which comes equipped with a monomorphism into the affine scheme $U \times U$) is separated. Hence we see that R is separated. In particular the morphisms s, t are separated as well as étale.

Since the composition $R \to U \times_S U \to U$ is locally of finite type we conclude that j is locally of finite type (see Morphisms, Lemma 01T8). As j is also a monomorphism it has finite fibres and we see that j is locally quasi-finite by Morphisms, Lemma 02NG. Altogether we see that j is separated and locally quasi-finite.

Our first step is to show that the quotient map $c : U \to F$ is representable. Consider a scheme T and a morphism $a : T \to F$. We have to show that the sheaf $G = T \times_{a,F,c} U$ is representable. As seen in the proofs of Lemmas 02WU and 02WV there exists an fppf covering $\{\varphi_i : T_i \to T\}_{i \in I}$ and morphisms $a_i : T_i \to U$ such that $a_i \times a_{i'} : T_i \times_T T_{i'} \to U \times_S U$ factors through R, and such that $c \circ a_i = a \circ \varphi_i$. As in the proof of Lemma 02WV we see that

$$\begin{aligned} T_i \times_{\varphi_i, T} G &= T_i \times_{\varphi_i, T} T \times_{a, U/R, c} U \\ &= T_i \times_{c \circ a_i, U/R, c} U \\ &= T_i \times_{a_i, U} U \times_{c, U/R, c} U \\ &= T_i \times_{a_i, U, t} R \end{aligned}$$

Since t is separated and étale, and in particular separated and locally quasi-finite (by Morphisms, Lemmas 02V5 and 02GV) we see that the restriction of G to each T_i is representable by a morphism of schemes $X_i \to T_i$ which is separated and locally quasi-finite. By Descent, Lemma 02W5 we obtain a descent datum $(X_i, \varphi_{ii'})$ relative to the fppf-covering $\{T_i \to T\}$. Since each $X_i \to T_i$ is separated and locally quasi-finite we see by More on Morphisms, Lemma 02W8 that this descent datum is

effective. Hence by Descent, Lemma 02W5 (2) we conclude that G is representable as desired.

The second step of the proof is to show that $U \to F$ is surjective and étale. This is clear from the above since in the first step above we saw that $G = T \times_{a,F,c} U$ is a scheme over T which base changes to schemes $X_i \to T_i$ which are surjective and étale. Thus $G \to T$ is surjective and étale (see Remark 02WH). Alternatively one can reread the proof of Lemma 02WV in the current situation.

The third and final step is to show that the diagonal map $F \to F \times F$ is representable. We first observe that the diagram

$$\begin{array}{ccc} R & \longrightarrow & F \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle \Delta} \\ U \times_S U & \longrightarrow & F \times F \end{array}$$

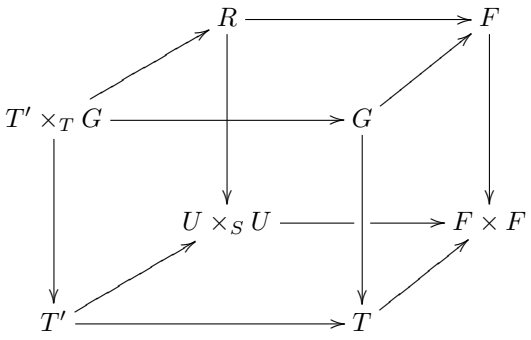
is a fibre product square. By Lemma 02WC the morphism $U \times_S U \to F \times F$ is representable (note that $h_U \times h_U = h_{U \times_S U}$). Moreover, by Lemma 02WM the morphism $U \times_S U \to F \times F$ is surjective and étale (note also that étale and surjective occur in the lists of Remarks 02WH and 02WG). It follows either from Lemma 02WB and the diagram above, or by writing $R \to F$ as $R \to U \to F$ and Lemmas 02W9 and 02WA that $R \to F$ is representable as well. Let T be a scheme and let $a : T \to F \times F$ be a morphism. We have to show that $G = T \times_{a,F \times F,\Delta} F$ is representable. By what was said above the morphism (of schemes)

$$T' = (U \times_S U) \times_{F \times F,a} T \longrightarrow T$$

is surjective and étale. Hence $\{T' \to T\}$ is an étale covering of T. Note also that

$$T' \times_T G = T' \times_{U \times_S U,j} R$$

as can be seen contemplating the following cube



Hence we see that the restriction of G to T' is representable by a scheme X, and moreover that the morphism $X \to T'$ is a base change of the morphism j. Hence $X \to T'$ is separated and locally quasi-finite (see second paragraph of the proof). By Descent, Lemma 02W5 we obtain a descent datum (X, φ) relative to the fppf-covering $\{T' \to T\}$. Since $X \to T'$ is separated and locally quasi-finite we see by More on Morphisms, Lemma 02W8 that this descent datum is effective. Hence by Descent, Lemma 02W5 (2) we conclude that G is representable as desired. □